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Neural-Network-Based Computational Framework for the Variational Theory of Lift

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We present a neural-network framework for modeling inviscid flows over airfoils, based on the philosophy of the Principle of Minimum Pressure Gradient (PMPG). In this setting, the fluid dynamics problem is recast as the minimization problem: among all kinematically admissible velocity fields, we seek the one that minimizes the L^2 norm of the pressure gradient. Neural networks serve as generic parametrizations of the admissible flow field, enabling a generic approximation of the infinite-dimensional function space over which the PMPG functional is minimized.

The neural network's inputs are the spatial coordinates and the output is the streamfunction to naturally generate a divergence-free velocity field. In this constrained optimization setup, boundary conditions such as no-penetration and far-field are imposed as well as equilibrium and irrotationality constraints. The resulting framework is pressure-free and data-free; relying solely on physical principles, the variational PMPG formulation, and the function-approximation capabilities of neural networks. We assess the accuracy of the proposed method on benchmark inviscid test cases and compare the neural-network solutions against classical potential-flow theory in the light of the variational theory of lift. Our results show that the framework recovers potential-flow velocity fields, captures circulation effects, and implicitly converges to Kutta-like conditions through the variational minimization. Although the present work focuses on steady inviscid airfoil problem, the formulation is, in principle, extensible to viscous flows, unsteady configurations, three-dimensional geometries, and more general rheological models within the same variational-neural framework.

I. Introduction

Classical incompressible aerodynamics over lifting bodies has traditionally been approached through potential-flow theory and conformal mapping, enabling closed-form solutions in idealized settings [1–5]. However, inviscid irrotational formulations admit a fundamental non-uniqueness: circulation is not determined by the governing Laplace equation alone, and additional closure assumptions are required to recover values of lift near the experimentally validated ones. The classical Kutta condition resolves this ambiguity by selecting the circulation Γ (and thus the lift $L = \rho U \Gamma$) through imposing finite-velocity flow at the sharp trailing edge [3, 6]. While effective for sharp edges, this closure becomes ambiguous for smooth or rounded trailing edges and has motivated long-standing questions about what truly determines circulation and lift in ideal-flow modeling [7–9].

To provide a generic closure that generalizes the sharp-edge special case, the variational theory of lift was introduced [10]. This theory is the special inviscid case of the Principle of Minimum Pressure Gradient (PMPG), which recasts incompressible flow as a minimization problem: among all kinematically admissible fields satisfying incompressibility and boundary conditions, nature selects the evolution of flow that minimizes a quadratic cost measuring the constraint force required to enforce continuity, namely the pressure gradient (or equivalently in inviscid flow, the norm of the convective acceleration) [11, 12]. This first principle is derived from Gauss's principle of least constraint in analytical mechanics [13, 14].

In the special case of potential flow over an airfoil, where the solution family is known up to a single free parameter Γ, the variational theory reduces to a one-dimensional minimization that picks the correct circulation Γ^* that minimizes the PMPG cost over the flow domain; which in this case is the curvature [10]. For general geometries and different flow regimes, however, PMPG prescribes *what* to minimize but does not prescribe an *ansatz* rich enough to represent

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admissible flows beyond a prioir known canonical forms. A practical computational framework therefore requires a generic parametrization of the admissible space of functions while retaining compatibility with the variational structure.

Recent advances in scientific machine learning, particularly Physics-Informed Neural Networks (PINNs), provide such parametrizations [15, 16]. Standard PINNs typically minimize the deviation between the model output and the data fed while also minimizing the governing PDE residuals. This prevents the domination of unphysical candidate solutions when training data is sparse or noisy.

While using a least squares cost function makes sense for PINNs, PMPG naturally suggests a physical loss functional. Building on recent PMPG-based data-free neural network solvers [17–22], we propose a PMPG–neural-network framework for inviscid airfoil flows in which the network parametrizes a streamfunction and is trained by minimizing the PMPG cost subject to kinematic constraints and boundary conditions.

This direction complements recent variational reformulations that cast incompressible dynamics as convex optimization and quadratic programming problems [23–25], and connects to PMPG-based ideal-flow solvers over arbitrary shapes using conformal mapping techniques [26]. In the present work, airfoil geometries and computational domains are generated using modified Zhukovsky conformal mapping, and the resulting PMPG-trained streamfunction model is validated against classical potential-flow predictions and the variational theory of lift.

The remainder of this paper is organized as follows. Section II reviews the theoretical foundations of PMPG as a variational formulation for fluid mechanics. Section III introduces the proposed PMPG-based neural-network framework, detailing the architecture and the physics imposed objective costs. Section IV showcases the implementation, including grid-generation using conformal mapping, boundary-condition imposition and Neural Network parameters. Section V presents numerical results and validation against potential-flow theory over a selected airfoil. Section VI concludes and outlines future research directions.

II. Fluid mechanics as a minimization problem

Consider an incompressible fluid of density ρ in a domain Ω with boundary $\partial\Omega$. The momentum equation may be written as

$$\rho\mathbf{a} \;=\; -\nabla p \;+\; \nabla\cdot\boldsymbol{\tau} \;+\; \mathbf{F}, \tag{1}$$

where $\mathbf{a} = \mathbf{u}_t + (\mathbf{u}\cdot\nabla)\mathbf{u}$ is the material acceleration, $\boldsymbol{\tau}$ is the viscous stress tensor (not necessarily restricted to Newtonian form), and $\mathbf{F}$ denotes external body forces. Incompressibility imposes the kinematic constraint of a divergence-free velocity field

$$\nabla\cdot\mathbf{u} = 0 \quad \text{in } \Omega, \tag{2}$$

together with the normal-velocity boundary condition (e.g., no-penetration on solid boundaries)

$$\mathbf{u}\cdot\mathbf{n} = 0 \quad \text{on } \partial\Omega. \tag{3}$$

Within this constrained dynamics, the pressure gradient ∇p acts as a *constraint force*: its role is to enforce Eq. (2) [11, 27].

A result of (Eq. (2) &(3)) is the orthogonality between divergence-free fields and gradients, which underlies projection methods for incompressible flows [27–29]. This viewpoint motivates a variational statement in which the pressure force is *minimized* subject to incompressibility and boundary conditions.

Gauss' principle of least constraint states that, at each instant, the acceleration of a constrained mechanical system minimizes the deviation from the acceleration the system would have in the absence of those constraints [13, 14]. Applying this philosophy to incompressible fluids leads to the Principle of Minimum Pressure Gradient (PMPG) [11].

Define the *free acceleration* as the acceleration induced by non-constraint forces,

$$\mathbf{a}_{\text{free}} \;=\; \frac{1}{\rho}\left(\nabla\cdot\boldsymbol{\tau} + \mathbf{F}\right). \tag{4}$$

Then Eq. (1) can be rewritten as

$$\mathbf{a} - \mathbf{a}_{\text{free}} \;=\; -\frac{1}{\rho}\nabla p, \tag{5}$$

showing explicitly that the pressure gradient is the measure of the *constraint force* needed to keep the motion incompressible. The PMPG asserts that the evolution of the flow selects the instantaneous acceleration (equivalently $\mathbf{u}_t$)

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that minimizes the Gauss-type functional

$$\mathcal{A}(\mathbf{u}_t) = \frac{1}{2}\int_\Omega \rho\ |\mathbf{a} - \mathbf{a}_{\text{free}}|^2\ d\mathbf{x}\ =\ \frac{1}{2\rho}\int_\Omega |\nabla p|^2\ d\mathbf{x}, \tag{6}$$

subject to kinematic admissibility (incompressibility and boundary compatibility). Taha *et al.*[11] proved that the first-order optimality conditions of Eq. (6) recovers the incompressible Navier–Stokes equations, establishing PMPG as a *minimum* principle for incompressible fluid mechanics (instantaneous minimization in acceleration space, not a stationary trajectory principle). The optimization variable is the admissible acceleration (or $\mathbf{u}_t$) and pressure emerges as the Lagrange multiplier associated with incompressibility.

In the inviscid, unforced case ($\boldsymbol{\tau} = \mathbf{0}$ and $\mathbf{F} = \mathbf{0}$), the free acceleration vanishes,and for steady ideal flows ($\mathbf{u}_t = \mathbf{0}$) , (6) reduces to

$$\mathcal{A} = \frac{\rho}{2}\int_\Omega |(\mathbf{u}\cdot\nabla)\mathbf{u}|^2\ d\mathbf{x}\ \equiv\ \frac{1}{2\rho}\int_\Omega |\nabla p|^2\ d\mathbf{x}, \tag{7}$$

it becomes a minimization of the domain-integrated squared convective acceleration [11]. This is considered to be a special case from Gauss' Principle of Least constraint reducing to Hertz's Principle of least curvature.

This form is particularly important for lifting flows. In classical potential-flow modeling around an airfoil, the admissible solution family is parameterized by the circulation Γ; Euler's equation alone does not determine Γ. The variational theory of lift provides an alternate closure to the Kutta condition by picking the circulation as the minimizer

$$\Gamma^*\ =\ \arg\min_{\Gamma}\ \mathcal{A}(\Gamma), \tag{8}$$

thereby generalizing the sharp-edge Kutta selection to a variational statement applicable beyond the classical special case [10, 11]. In the present work, we exploit Eq. (7) & (8) by replacing the finite-dimensional ansatz $\mathbf{u}(\cdot;\Gamma)$ with a neural network parametrization, while keeping the PMPG cost as the governing objective to minimize. The flow field is parametrized by a neural network with trainable parameters $\boldsymbol{\theta}$:

$$\mathbf{u}(\mathbf{x})\ =\ \mathcal{N}\mathcal{N}(\mathbf{x};\boldsymbol{\theta}), \tag{9}$$

The problem is then transformed into finding the minimizer $\boldsymbol{\theta}^*$

$$\boldsymbol{\theta}^*\ =\ \arg\min_{\boldsymbol{\theta}}\ \mathcal{A}(\boldsymbol{\theta}), \tag{10}$$

while satisfying the continuity constraint $\nabla\cdot\mathbf{u} = 0$, the no-penetration boundary condition $\mathbf{u}\cdot\mathbf{n} = 0$ on boundary $\partial\Omega$ and the far-field boundary condition.

III. PMPG-based Neural Network framework

In this work we adopt a data-free, physics-constrained approach that leverages PMPG to guide the training of a neural-network model for incompressible flow over an airfoil. Following the methodology in previous works [17, 18, 21], the network is trained not by minimizing PDE residuals, but by minimizing the variational PMPG cost functional grounded in analytical mechanics.

The integration of PMPG with neural parametrizations has already demonstrated strong performance across multiple benchmark regimes, including flow over an inviscid cylinder [17–20], lid-driven cavity flow [20, 21], rotating cylinders [22], and Poiseuille-type flows [20]. Figure 1 showcases a qualitative analysis of the velocity field quivers generated by the PMPG-based Neural Network solvers and analytical solutions in two problems; the rotating cylinder problem and the inviscid flow over a cylinder. PMPG-based Neural Network solvers gave almost identical results compared to the well-known flow over cylinder and the variational work by Shorbagy and Taha [30].

Computation-wise, PMPG-based solvers have an edge over classical PINN formulations due to the elimination of the need to train a separate pressure network model [17, 18, 21].

Let $\Omega \subset \mathbb{R}^2$ denote the flow domain with boundary $\partial\Omega$, and let $\mathbf{u}(\mathbf{x})$ be the steady velocity field to be determined. In the steady inviscid setting considered here, PMPG reduces to minimizing the domain-integrated squared convective acceleration (equivalently the squared pressure-gradient magnitude) [10, 11]. We therefore pose the constrained

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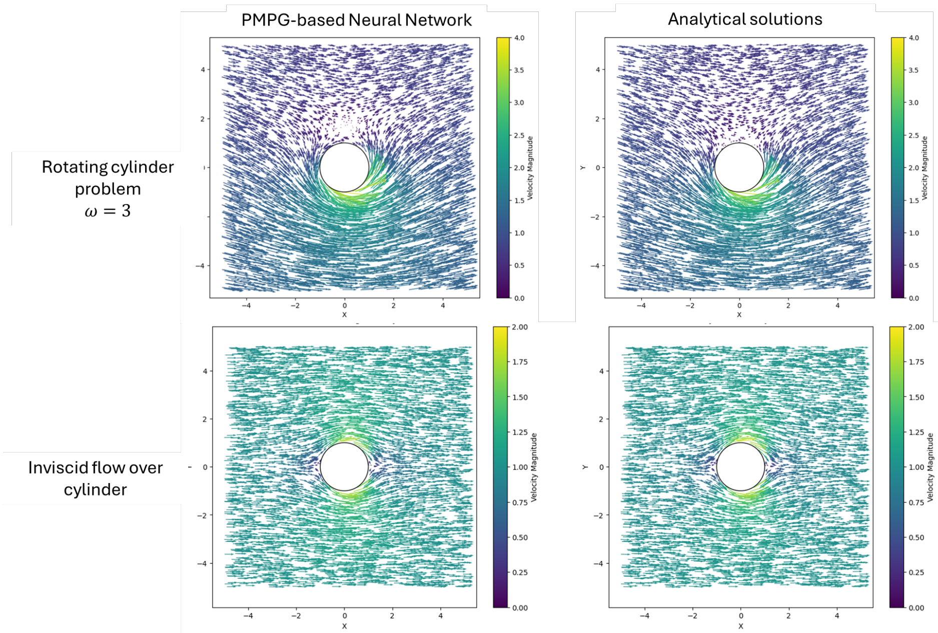

Fig. 1 Previous works comparing the velocity quivers of PMPG-based Neural Network solvers for the flow over rotating cylinder at non-dimensional rotation $\omega = 3$ [31] and the inviscid flow over cylinder [17, 18] with their respective analytical solutions.

minimization problem

$$\begin{aligned}
\min_{\mathbf{u}(\mathbf{x};\boldsymbol{\theta})} \quad & \mathcal{A}(\boldsymbol{\theta}) = \frac{\rho}{2}\int_{\Omega} |(\mathbf{u}\cdot\nabla)\mathbf{u}|^2 \, d\mathbf{x} & \\
\text{s.t.} \quad & \nabla\cdot\mathbf{u} = 0 & \text{in } \Omega, \\
& \nabla\times\big((\mathbf{u}\cdot\nabla)\mathbf{u}\big) = 0 & \text{in } \Omega, \\
& \nabla\times\mathbf{u} = 0 & \text{in } \Omega, \\
& \mathbf{u}\cdot\mathbf{n} = 0 & \text{on } \partial\Omega, \\
& \mathbf{u}(\mathbf{x}) \longrightarrow \mathbf{u}_\infty(\mathbf{x};\Gamma) & \text{as } r\longrightarrow\infty,
\end{aligned} \tag{11}$$

where $\boldsymbol{\theta}$ denotes the parameters of the chosen function class (here, a neural-network parametrization), $\partial\Omega$ is the airfoil boundary. The far-field target is chosen as a uniform stream at angle of attack α superimposed by a decaying vortex contribution with circulation Γ. In polar coordinates (r,θ) this target may be written as

$$u_r^\infty(r,\theta;\Gamma) = U_\infty\left(1-\frac{R^2}{r^2}\right)\cos(\theta-\alpha), \qquad u_\theta^\infty(r,\theta;\Gamma) = -U_\infty\left(1+\frac{R^2}{r^2}\right)\sin(\theta-\alpha) - \frac{\Gamma}{2\pi r}, \tag{12}$$

.

The quantity minimized in Eq. (11) is the squared norm of the convective acceleration integrated over the domain. The constraints enforce: (a) incompressibility through $\nabla\cdot\mathbf{u} = 0$; (b) a curl-free convective-acceleration condition $\nabla\times((\mathbf{u}\cdot\nabla)\mathbf{u}) = 0$, reflecting the steady Euler structure $(\mathbf{u}\cdot\nabla)\mathbf{u} = -(1/\rho)\nabla p$; (c) irrotationality $\nabla\times\mathbf{u} = 0$, restricting the solution class to potential-flow states consistent with the ideal-flow regime targeted in this work; (d) the no-penetration boundary condition $\mathbf{u}\cdot\mathbf{n} = 0$ on the airfoil boundary $\partial\Omega$; and (e) the far-field condition $\mathbf{u}\to\mathbf{u}_\infty(\cdot;\Gamma)$, which anchors the solution on the truncated domain and allows circulation to be freely selected by the minimization process.

To obtain a flexible approximation space while retaining exact incompressibility in two dimensions, we use a streamfunction as the output of the neural network parametrized by $\boldsymbol{\theta}$,

$$\psi(\mathbf{x}) \;=\; \mathrm{NN}(\mathbf{x};\boldsymbol{\theta}), \qquad \mathbf{u}(\mathbf{x}) \;=\; \nabla^\perp\psi(\mathbf{x}) \;=\; \begin{bmatrix}\partial\psi/\partial y \\ -\partial\psi/\partial x\end{bmatrix}. \tag{13}$$

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This construction satisfies the continuity equation $\nabla \cdot \mathbf{u} = 0$ naturally, thereby eliminating the continuity constraint from the minimization problem. In 3D setups a vector potential structure may be used to naturally satisfy incompressibility.

The PMPG objective defines variational cost, but the admissible solution set is characterized by constraints (e.g., no-penetration on boundary, far-field behavior, irrotationality, and equilibrium) that must be satisfied to obtain a flow that follows the governing equations. The main cost term drives the model toward the PMPG-optimal state, while the constraint terms delimit the feasible manifold and must be enforced to within a prescribed tolerance. Among several approaches for handling equality constraints in neural training—including augmented Lagrangian strategies [32]—we adopt a penalty method due to its simplicity and seamless integration into standard optimization workflows. The penalty method converts the constrained minimization into an unconstrained one by augmenting the objective with terms that quantify constraint violations, assigning higher weights to terms that require near-exact satisfaction by penalizing constraint violations [33, 34]. Concretely, for a neural parametrization $\mathbf{u}_{\boldsymbol{\theta}}$, we minimize a weighted total loss of the form

$$\mathcal{L}_{\text{tot}}(\boldsymbol{\theta}) = \mathcal{L}_{\text{PMPG}}(\boldsymbol{\theta}) + \sum_m \lambda_m \, \mathcal{L}_m(\boldsymbol{\theta}), \tag{14}$$

where $\mathcal{L}_{\text{PMPG}}$ is the PMPG functional and each $\mathcal{L}_m$ is a nonnegative penalty (typically quadratic) that vanishes when the corresponding constraint is satisfied. In this work we use quadratic penalties on (i) the no-penetration boundary condition, (ii) the far-field condition, (iii) a curl-free convective-acceleration condition $\nabla \times ((\mathbf{u} \cdot \nabla)\mathbf{u}) = 0$, (iv) irrotationality of the velocity field. The associated weights λ_m are chosen sufficiently large enough to drive the root-mean-square constraint residuals below a prescribed tolerance ϵ.

The primary term in the loss function is the magnitude of the pressure gradient; for steady, inviscid flow this is equivalent to the L^2-norm of the convective acceleration:

$$\mathcal{A} = \frac{1}{2} \int_{\Omega} |(\mathbf{u} \cdot \nabla)\mathbf{u}|^2 \; d\mathbf{x}. \tag{15}$$

In practice, the integral in (15) is approximated using interior collocation points $\{\mathbf{x}_i\}_{i=1}^{N}$:

$$\mathcal{L}_{PMPG} = \sum_{i=1}^{N} |(\mathbf{u} \cdot \nabla)\mathbf{u}|^2 \, (\mathbf{x}_i). \tag{16}$$

The equilibrium constraint is enforced through the equilibrium loss, $\mathcal{L}_{Eqm.}$, which penalizes the curl of the convective acceleration:

$$\mathcal{L}_{Eqm.} = \frac{1}{N} \sum_{i=1}^{N} \left| \left(\nabla \times \left((\mathbf{u} \cdot \nabla)\mathbf{u}\right)\right)_z \right|^2 (\mathbf{x}_i). \tag{17}$$

The boundary conditions are enforced similarly. The no-penetration condition on the airfoil boundary $\partial\Omega$ is imposed by penalizing the normal velocity:

$$\mathcal{L}_{BC} = \frac{1}{N^{BC}} \sum_{j=1}^{N^{BC}} |\mathbf{u} \cdot \mathbf{n}|^2 \, (\mathbf{x}_j^{BC}), \qquad \mathbf{x}_j^{BC} \in \partial\Omega, \tag{18}$$

where $\mathbf{n}$ denotes the outward unit normal on $\partial\Omega$.

The far-field condition is enforced through the far-field loss $\mathcal{L}_{FF}$ on a far boundary using collocation points $\{\mathbf{x}_k^{FF}\}_{k=1}^{N^{FF}}$:

$$\mathcal{L}_{FF} = \frac{1}{N^{FF}} \sum_{k=1}^{N^{FF}} \left| \mathbf{u}(\mathbf{x}_k^{FF}) - \mathbf{u}_\infty(\mathbf{x}_k^{FF}; \Gamma) \right|^2, \tag{19}$$

where $\mathbf{u}_\infty(\cdot; \Gamma)$ is the prescribed freestream target corrected by the computed circulation Γ on the airfoil.

Finally, to restrict the solution to the potential-flow class in the interior, we enforce irrotationality via the irrotationality loss $\mathcal{L}_{\text{Irrot.}}$:

$$\mathcal{L}_{\text{Irrot.}} = \frac{1}{N} \sum_{i=1}^{N} \left| (\nabla \times \mathbf{u})_z \right|^2 (\mathbf{x}_i). \tag{20}$$

All terms are combined into the total loss

$$\mathcal{L}_{total} = \lambda_{\text{PMPG.}} \mathcal{L}_{\text{PMPG.}} + \lambda_{\text{Eqm.}} \mathcal{L}_{\text{Eqm.}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_{\text{FF}} \mathcal{L}_{\text{FF}} + \lambda_{\text{Irrot.}} \mathcal{L}_{\text{Irrot.}}, \tag{21}$$

where $\lambda_{(\cdot)} > 0$ are penalty weights selected to enforce constraint satisfaction within a prescribed tolerance.

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IV. Implementation

We follow the modified Zhukovsky mapping used by Gonzalez and Taha [10], which maps a circular cylinder in the auxiliary ζ-plane to an airfoil boundary in the physical z-plane:

$$z(\zeta) = \zeta + \mu + \Delta \frac{C^2}{\zeta + \mu}, \qquad \Delta = \frac{1 - D}{1 + D}. \tag{22}$$

Here, μ offsets the cylinder center (controlling camber and thickness), C sets the overall geometric scale, and $D \in [0, 1]$ controls trailing-edge smoothness ($D = 0$ yields a sharp trailing edge, while $D = 1$ recovers a circular cylinder). Airfoil shapes are reported using the pair (τ, D), where τ denotes a thickness-to-chord measure; for example, $\tau = 0.1$ corresponds to an airfoil with approximately 10% relative thickness.

For convenience, we parameterize the map coefficients (μ, C) using τ through an intermediate eccentricity-like quantity ε:

$$\varepsilon = \frac{\tau}{(3\sqrt{3})/4}, \qquad C = \frac{1}{1 + \varepsilon}, \qquad \mu = \varepsilon\, C, \qquad \Delta = \frac{1 - D}{1 + D}. \tag{23}$$

In this work we use this in the grid generation in the airfoil domain, where points in the cylindrical ζ -domain are conformed to a z-domain grid points corresponding to an airfoil [Figure 2]. This includes (i) the collocation points on which the PMPG cost is computed and the curl of convective acceleration and irrotationality constraints are satisfied, (ii) the boundary points to satisfy the no penetration, and (iii) the far-field points on which the far-field condition is imposed.

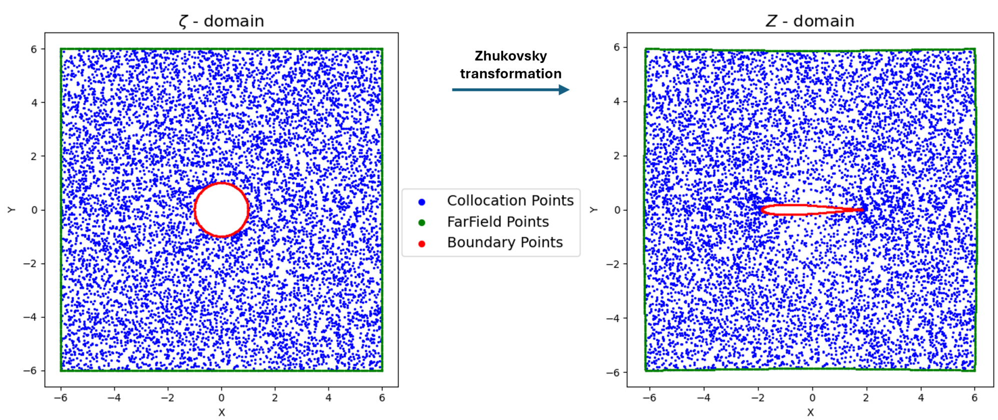

Fig. 2 Transformed collocation points. Blue points refer to the domain points, green points refer to the far-field points and red points refer to the boundary points.

The number of collocation points to generate the grid in the work is 9000 randomly sampled points, with a cylinder radius of unity and domain size of 6×6. The number of Boundary points used to impose the no-penetration boundary condition is 200 and the number of points for far-field boundary condition is 800.

In the cylindrical domain, the velocity field $\mathbf{u}$ in theory can be derived from the complex potential $F(\zeta)$, constructed as:

$$F(\zeta) = U_\infty \left(e^{-i\alpha}\zeta + e^{i\alpha}\frac{R^2}{\zeta} \right) - \frac{i\Gamma}{2\pi}\log \zeta \tag{24}$$

The complex velocity W can be computed as:

$$W(\zeta) = \frac{dF}{d\zeta} = u_x - iu_y. \tag{25}$$

In previous analytical works [10], the circulation Γ is treated as a free parameter, and the family of admissible flow fields $\mathbf{u}(\zeta; \Gamma)$ satisfies the no-penetration boundary condition by construction. However, Γ is not determined by kinematics alone. Instead, minimizing the convective acceleration cost Eq. 7 determines the correct circulation Γ^*. In this work we open the design space to parametrize the flow based on the neural network weights and biases θ_{NN} and don't confine the solver to sweep in the family of solutions with the form of Equations 24 & 25.

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However, we use this form in imposing the far-field boundary condition, where the $\Gamma = \oint_C \mathbf{u} \cdot d\boldsymbol{\ell}$ is calculated on the airfoil coupling the far-field with the parameter of circulation on the airfoil as mentioned in Eq. 12. For a conformal map $z = \mathcal{F}(\zeta)$, this can be achieved by conforming the form mentioned in Eq. 12 by defining complex velocity in the z domain using the chain rule

$$W_z(z;\Gamma) = W(\zeta)\frac{d\zeta}{dz} \tag{26}$$

the far-field velocity in the airfoil plane is given by

$$\mathbf{u}^{\infty}(z;\Gamma) = \begin{bmatrix} u_x^{\infty}(z;\Gamma) \\ u_y^{\infty}(z;\Gamma) \end{bmatrix} = \begin{bmatrix} \Re\big(W_z(z;\Gamma)\big) \\ -\Im\big(W_z(z;\Gamma)\big) \end{bmatrix}, \qquad \text{on far-field points} \tag{27}$$

All optimization variables and network evaluations are performed in the z-domain; the ζ-domain appears only through the mapping operators used to generate geometry and evaluate analytic far-field targets.

Following the PMPG-based neural network solvers in our previous works [17, 18, 21], we represent the two-dimensional incompressible velocity field through a neural streamfunction ψ. The network takes Cartesian coordinates $\mathbf{x} = (x, y) \in \mathbb{R}^2$ as inputs and returns a single scalar output,

$$\psi(x, y) = \mathcal{NN}\big([x, y]; \boldsymbol{\theta}\big), \tag{28}$$

In this work we use a fully-connected multilayer perceptron with two hidden layers of width 50 neurons each and a $\tanh(\cdot)$ activation function. Neural Network parameters $\boldsymbol{\theta}$ are optimized by minimizing the total loss (Eq. (21)) using the Adam optimizer [35]. Figure 3 shows the Neural Network architecture, the flow of information for training as well as the loss functions. Gradient calculations for ψ and $\mathbf{u}$ needed for the calculation of vorticity $\nabla \times \mathbf{u}$ and convective acceleration $(\mathbf{u} \cdot \nabla)\mathbf{u}$ and its curl are computed using automatic differentiation.

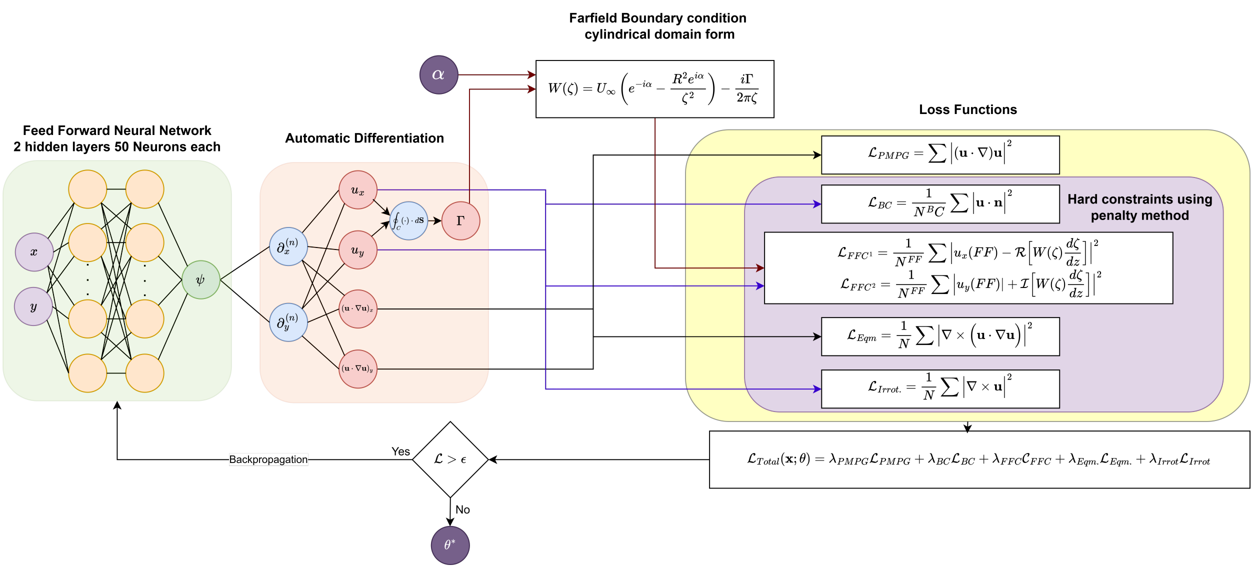

Fig. 3 Schematic of the PMPG-based Neural Network with loss functions

V. Results and discussion

We validate the proposed framework by comparing the steady inviscid flow over a mapped airfoil ($\tau = 0.1$, $D = 0.01$, $\alpha = 2°$) against the corresponding analytical potential-flow solution with the Γ^* calculated using the work of Gonzalez and Taha [10]. Figures 4–5 show that the streamfunction contours and velocity-field quivers closely match the analytical reference.

Table 1 provides a quantitative assessment of accuracy and constraint satisfaction. The PMPG–NN solution achieves a low velocity error (3.4×10^{-2}) while maintaining near machine-precision divergence-free velocity field

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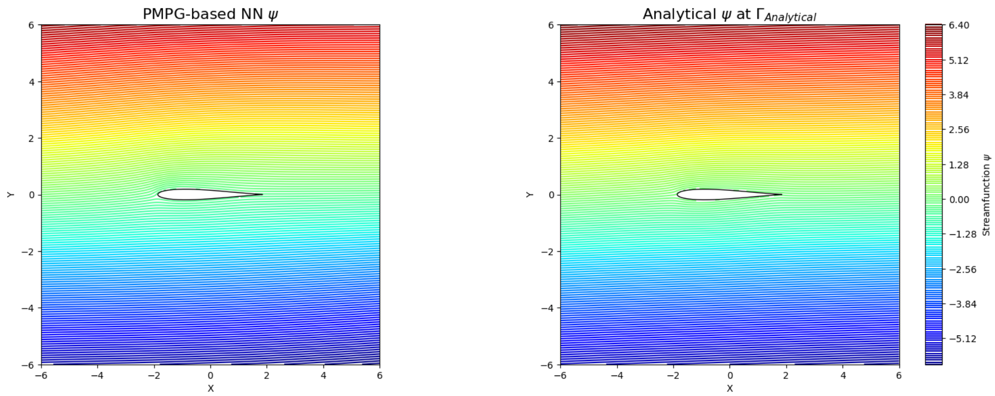

Fig. 4 Streamfunction comparison between the model generated flow and the analytical flow using $\Gamma_{Analytical}$ at airfoil of parameters $\tau = 0.1, D = 0.01, \alpha = 2°$.

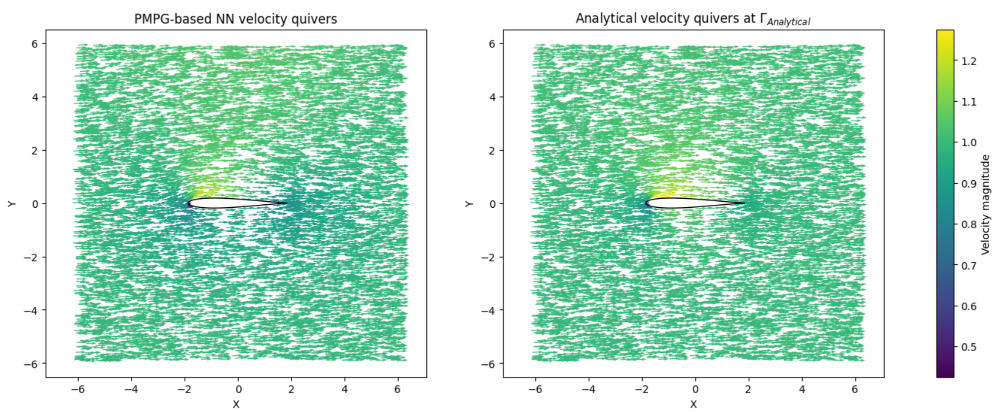

Fig. 5 Velocity quivers comparison between the model generated flow and the analytical flow using $\Gamma_{Analytical}$ at airfoil of parameters $\tau = 0.1, D = 0.01, \alpha = 2°$.

($\|\nabla \cdot \mathbf{u}\| = 1.93 \times 10^{-7}$). This is expected due to the satisfaction of the incompressibility constraint by the streamfunction construction. The no-penetration and far-field constraints are satisfied to within $O(10^{-3})$ and $O(10^{-2})$, respectively, indicating that the boundary and far-field conditions are strongly enforced by the penalty method formulation. The norm of generated vorticity and curl of convective acceleration remain negligible ($O(10^{-2})$) showcasing a good satisfaction of both constraints imposed on the velocity and the convective acceleration fields. The integrated domain cost also quiet matches the analytical calculation with a value of 0.6159 vs 0.5828. The computed sectional lift coefficient $C_L = \frac{2\Gamma}{U_\infty c}$ yields $C_L = 0.2636$ for the PMPG–based Neural Network solution and $C_L = 0.2313$ for the analytical circulation Γ^* reported by Gonzalez and Taha [10] (with $U_\infty = 1$ and $c = 4$). Thin-airfoil theory predicts

$$C_L \approx 2\pi\alpha, \tag{29}$$

where at $\alpha = 2°$ ($\alpha \simeq 0.0349$ rad) one expects $C_L \approx 0.219$, consistent in magnitude with both values. It is obvious the over-estimation of both the PMPG cost and the C_L using our model, which can be attributed to sparse point distribution especially around the airfoil geomtery. This gap can be narrowed down by refining the points around the airfoil with higher density. As a validation proof of concept, the results are acceptable in order of magnitude and the basic flow properties are qualitatively captured as illustrated in Figures 4 and 5.

VI. Concluding remarks and future insights

This work presented a data-free, variational neural framework for steady incompressible flow over airfoils, built on the Principle of Minimum Pressure Gradient (PMPG). The framework works by minimizing the domain-integrated

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Table 1 Quantitative comparison between the PMPG-based neural network solution and the analytical potential-flow solution using Γ^* from [10] for $\tau = 0.1$, $D = 0.01$, $\alpha = 2^\circ$ and chord length $c = 4$.

Metric	PMPG–NN	Analytical
PMPG cost	0.6159	0.5828
Circulation Γ	0.5273	0.4625
Coefficient of Lift $C_L = \frac{2\Gamma}{U_\infty c}$	0.2636	0.2313
Velocity Error Norm $\|\mathbf{u} - \mathbf{u}_{\text{analytical}}\|$	3.40×10^{-2}	0
Curl of convective acceleration Norm $\|(\mathbf{u} \cdot \nabla)\mathbf{u}\|$	1.32×10^{-2}	0
Vorticity norm $\|(\nabla \times \mathbf{u}\|$	1.63×10^{-2}	0
No-penetration residual norm	3.60×10^{-3}	0
Far-field residual norm	8.78×10^{-3}	0
Divergence norm $\|\nabla \cdot \mathbf{u}\|$	1.928×10^{-07}	0

squared convective acceleration (curvature) while enforcing no penetration boundary condition, far-field boundary condition, incompressibility, equilibrium and irrotationality. For a representative mapped airfoil case as a validation proof of concept ($\tau = 0.1$, $D = 0.01$, $\alpha = 2^\circ$), the modeled streamfunction and velocity field closely match the analytical potential-flow reference, while achieving low constraint residuals and a coefficient of lift C_L comparable to the analytically obtained values reported in the variational lift theory.

Several directions can extend and strengthen the present framework. First, a systematic sensitivity study of the circulation selection with respect to domain truncation, collocation density, and penalty weights would clarify the robustness of the variational closure in practical discretizations. Second, sweeping the solver over different airfoils and angle of attacks to find limitations of the solver, especially at rounded trailing edges and thicker airfoils. Third, a study on the effect of removing the irrotationality constraint where separation for thicker airfoils and at higher angle of attacks are expected. A glimpse of those separating flows has been observed in the authors' pilot simulations which will be discussed in future works. Fourth, extending the formulation to viscous and unsteady regimes using the general PMPG principle (instantaneous minimization in acceleration space) would enable data-free solvers for Navier–Stokes flows. Finally, the proposed PMPG-based solver can be naturally embedded within an airfoil shape-optimization loop, where the geometry parameters (or a higher-dimensional shape representation) are treated as design variables and updated to improve aerodynamic performance. This opens the door to gradient-enabled optimization for increasing lift, reducing drag and exploring innovative geometries beyond classical parameterizations, while retaining a physics-grounded variational objective.

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